

Frequency-Domain Modeling of Interconnects Based on Assemble Neural Network for 3-D Integration

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Abstract—This article proposes a novel neural network (NN) architecture combining convolutional and transposed convolutional NNs (TCNNs) to accurately and efficiently model the S -parameter of interconnects for 3-D integration. The network incorporates physical consistency constraints, specifically causality and passivity, into its design to ensure the physical effectiveness of the output. The transposed convolutional network serves as a subnetwork to map the relationship between the geometrical parameters and the S -parameter for substructures. Then, the S -parameters of individual substructures are cascaded to deal with a complex structure composed of substructures. A coupling NN (CONN), with causality and passivity constraints, is developed to map the coarse cascaded S -parameters to the fine, accurate S -parameters. With the help of this high-dimensional space mapping, a small amount of electromagnetic (EM) simulation data of complex interconnect structures is sufficient to learn the relationship between cascaded and real S -parameters. To ensure the completeness of the training set distribution when training the CONN on small datasets, a sensitivity analysis-based training set screening method is proposed to enhance the training performance of the CONN. The proposed algorithm is demonstrated in two different assemble structure applications. The results highlight the effectiveness, flexibility, and versatility of the proposed architecture in modeling complex structures with small, costly simulation data while maintaining accuracy and physical consistency.

Index Terms—Assemble neural network (NN), electromagnetic (EM) modeling, interconnect, neuro-space mapping, physical consistency, sensitivity analysis (SA), S -parameter.

I. INTRODUCTION

THE advancement of semiconductor processes is increasingly restricted as the feature size of the transistor approaches the physical limits, posing significant challenges to

the continuation of Moore's Law. Three-dimensional integration technology offers an effective way to overcome this bottleneck. The effectiveness of this technology largely relies on the performance of high-speed signal transmission channels both between and within the chips. Rapid analysis and design of systems with high-speed, high-density interconnects remains a challenging task due to high signal propagation speeds and increasingly compact footprint of interconnect structures. Traditional modeling of interconnects and passive components typically relies on commercial electromagnetic (EM) simulation software, which provides accurate iterative numerical solutions to partial differential equations for targeted 3-D structures. However, as structures increase in complexity, especially when composed of multiple fundamental components, this approach becomes time-intensive and demands substantial computational resources.

To date, numerous studies have taken the distributed RLGC to accurately estimate on-chip interconnection performance [1], [2], [3], [4], [5], [6], [7]. Considering the discontinuities at interfaces between different components and the increased complexity of multicomponent systems, the equivalent circuit model alone falls short in accurately characterizing these highly complex structures. Efficiently modeling and analyzing interconnect structures composed of diverse components remains a challenging task. With advancements in machine learning (ML), particularly neural networks (NNs), it is now possible to construct rapid and accurate surrogate models tailored for these tasks through a single training process. Various ML models have been proved effective and powerful for characterization of the nonlinear relationship between design parameter space and response parameter space for filters, [8], [9], [10], [11], interconnects [12], [13], [14], [15], [16], and advanced packaging [17], [18], [19]. Although ML techniques have demonstrated significant advantages across various tasks, they still face limitations in terms of model reusability and the high cost associated with acquiring complex datasets for training.

ML is inherently a data-driven approach, which means that its performance is largely contingent on the quantity and quality of the datasets used for training. In high-frequency and high-speed circuit design, the increasing parasitic effects of interconnect structures render the acquisition of EM simulation data costly, particularly for complex interconnects composed of various substructures. To address this issue, techniques such as data clustering, sensitivity analysis (SA), and feature extraction are extensively employed to optimize the use of these expensive datasets. Novel network architectures have

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also been proposed in this field such as the knowledge-based NN [20], [21], hybrid NN [22], [23], transfer learning [24], [25], [26], generative adversarial networks [27], [28], Graph NN [29], and physics-informed NN [30], [31], [32], to accelerate network convergence and mitigate data dependence.

Despite these advancements, one significant challenge remains that both the dataset and the well-trained network are often task-specific. An NN that is well-trained for a particular task tends to perform optimally only for the specific task. Its performance may not transfer well to other tasks, which means that the network lacks reusability. To enhance the generality of the network, one approach is to construct a massive network with billions of neurons and learnable parameters, trained on as extensive a dataset as possible, like large-scale language models like ChatGPT [33]. Obviously, this approach prevents waste of the dataset, but it increases the demand for even larger datasets. Another strategy is to decompose a complex task into simpler foundational tasks, train separate networks for each task, and these foundational tasks can then be combined to address more complex tasks, enabling the reuse of the networks trained for the basic tasks across different application scenarios. There has been ongoing discussion in the literature regarding the use of multiple networks to solve problems more efficiently [34], [35], [36], but these techniques all aim to complete a single task, which means the networks they use are not reusable. Transfer learning and knowledge distillation techniques [37], [38] can partially address the challenge of improving the reusability of models to new tasks. However, they still suffer from limited explainability and a lack of transparency regarding the underlying physical mechanisms.

To solve this problem, a novel multiple NN architecture named assemble NN for modeling S -parameters of interconnect structures is proposed in this article. For traditional surrogate modeling methods, the task goal is often to directly construct a surrogate model for an entire system or structure, which is like “one model for one task.” As for the model proposed in this article, the complex task is divided into basic tasks that can be used to compose other complex tasks, and then basic surrogate models are set up for the basic tasks. These basic surrogate models, named Reusable Lego NNs, can be used for different tasks, which is like “one model for multiple tasks.” For modeling the S -parameter of complex interconnects, different reusable Lego NNs (RLNNs) for basic components that can compose various complex interconnect structures are trained well, and the results from the basic networks are cascaded according to the sequence of the basic components for the target interconnect. A coupling NN (CONN) is then introduced to establish a space mapping from the coarse cascaded S -parameters to the accurate S -parameters, thereby mitigating the impact of coupling effects between different basic components. To make sure that the S -parameters output from the CONN satisfy the physical consistency, hard constraints [39] are added into the network to ensure the causality and passivity of the S -parameter. Since the completeness of the training set distribution plays a critical role in determining the performance of network training on small-sample datasets, an SA-based training set screening method is proposed to enhance the training effectiveness of the CONN.

The proposed architecture demonstrates significant superiority in the applications of two types of interconnect structures.

The main contribution could be summarized as follows.

1) The trained basic RLNNs can be used repeatedly to solve different problems, avoiding the waste of expensive datasets and network training.

2) Decomposing the surrogate model of complex structure into a combination of surrogate models of simple structures, avoiding the acquisition of large amounts of expensive datasets and saving computing resources and the dataset acquisition time.

3) A high-dimensional to high-dimensional CONN architecture is proposed to achieve the mapping from the coarse model to the fine model with a small amount of data.

4) An SA-based training set screening method is proposed to further enhance the performance of the CONN for training on small-sample datasets.

The rest of this article is organized as follows. Section II presents the proposed assemble NN in detail and introduces the method to implement hard constraints of physical consistency of S -parameters on the NN training. Section III introduces the SA-based training set screening method for performance enhancement of the CONN. The applications of the proposed algorithm to complex interconnect structures are presented in Sections IV and V. In Section IV, the interconnect structure composed of the substrate integrated coaxial line (SICL) and the ball grid array (BGA) is modeled, while the 3-D interconnect, including the through-silicon via (TSV) and the redistribution line (RDL), is investigated in Section V. Some conclusions are finally drawn in Section VII.

II. PROPOSED ASSEMBLE NN FOR MULTICOMPONENT STRUCTURE

A. Framework of Assemble NN

The proposed assemble NN consists of three blocks: an RLNN for each single component, an S -parameter cascading model (SCM), an SA-based training set screening method, and a CONN. The complete architecture of the proposed assemble model is shown in Fig. 1. When the entire architecture is implemented, RLNNs are used as a surrogate model to generate S -parameters of each component separately, and the number depends on how many components the system consists of. Then, the SCM module cascades the output of each RLNN in sequence through the $ABCD$ matrix. Then, an SA-based training set screening method is proposed to guide the training set selection of the CONN. The CONN is finally used to learn the space mapping relationship between coarse cascaded S -parameter results and the fine, accurate S -parameter results. The SA method serves as an additional component to enhance the performance of the CONN without affecting the completeness of the overall NN structure. This method will be discussed separately in Section III.

B. Reusable Lego NN

The RLNN is an indispensable cornerstone of the whole assemble NN. When the application is determined, it will be divided into several parts, whose number depends on how many subcomponents the application has. Since the subcomponents should be indivisible and the most basic blocks that build up the whole application system, the network used is the so-called RLNN. Once the RLNNs are trained well, the basic blocks they represent can be used to form different assemble structures. This usage is like stacking Lego blocks, which is why the RLNN got its name, as illustrated in Fig. 2.

The specific task of the RLNN is learning the mapping. The input and output of the network are the design parameter

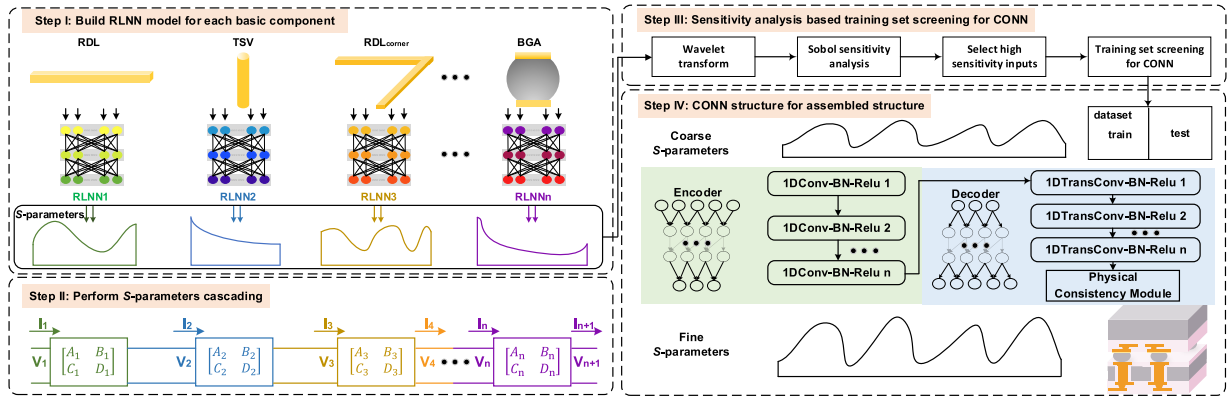


Fig. 1. Complete architecture of the proposed assemble model.

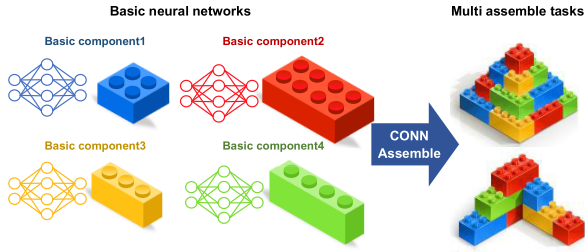


Fig. 2. Stacking RLNNs like Lego blocks to complete different assemble tasks.

space and the S -parameters of the divided subcomponents. Taking a two-port element as an example, we assume that the number of the design parameters is n , the input vector of the network is $\tilde{x} = [x_1, x_2, x_3, \dots, x_n]^T$, and the S -parameter matrix corresponding to each input vector $\tilde{x}$ is as follows:

$$\tilde{S} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \quad (1)$$

where $S_{ij} = \text{Re}S_{ij} + j\text{Im}S_{ij}$ ($i, j = [1, 2]$), and $\text{Re}S_{ij}$ and $\text{Im}S_{ij}$ denote the real and imaginary parts of S_{ij} . To facilitate the network training, the real and imaginary curves of the S -parameter are split. When the S -parameter network satisfies reciprocity (i.e., $S_{ij} = S_{ji}$), only S_{11} and S_{21} are selected for training, as the remaining elements can be inferred accordingly. If the number of samples at the S -parameter frequency point is m , the detailed S -parameter matrix corresponding to the geometrical parameter mapping is as follows:

$$y_{\text{RLNN}} = [\text{Re}(S_{ij}), \text{Im}(S_{ij})], \quad \text{for } i \leq j. \quad (2)$$

When the S -parameter network does not satisfy reciprocity (i.e., $S_{ij} \neq S_{ji}$), all relevant S -parameter elements must be included explicitly in the training process. In this case, both forward and reverse transmission/reflection characteristics (e.g., S_{11}, S_{12}, S_{21} , and S_{22}) are treated as independent quantities and modeled separately. This ensures that the NN captures the full directional behavior of the system, which is especially critical in the presence of asymmetric structures and nonuniform tapered lines. In this case, the output of each RLNN is defined as

$$\widetilde{y_{\text{RLNN}}} = [\text{Re}(S_{ij}), \text{Im}(S_{ij})], \quad i \neq j \text{ or } i = j. \quad (3)$$

In practical applications, the subelement after block division is relatively simple, and the dimension n of the design parameter space is often in a single digit. When obtaining S -parameters, the sampling point m is often between 100 and 1000 in the specified frequency range. The mapping from $\tilde{x}$ to $\tilde{S}$

that the network needs to learn is the mapping from n -dimensional column vector to a large matrix with size $4 \times m$, that is, the mapping from low-dimensional space to high-dimensional space.

In the overall assemble NN architecture, the RLNN module refers to a class of networks designed for accurately mapping the geometric parameters of passive devices to the S -parameter space, like the deep NN (DNN), support vector machine (SVM), and other related methods. It also serves as a plug-and-play module. As an illustrative example, the spectral transposed convolutional NN (TCNN) [40] is utilized as the RLNN module. Compared with the traditional convolutional NN (CNN), the TCNN can be regarded as the inverse operation of the CNN. As is well known, the CNN has been widely used in various fields, including image recognition, target detection, and image generation [41]. The most common task of CNNs is to extract features from images through convolution kernels. The essence of this task is to regard images as high-dimensional numerical matrices, black-and-white images as 2-D matrices, and color images as 3-D matrices. Then, the information contained therein is compressed and integrated into low-dimensional features. The TCNN uses the inverse operation of the CNN to achieve upsampling from a low-dimensional space to a high-dimensional space, such as restoring the size of the image from a small resolution to a large resolution. The S -parameter matrix, the target of this article's task, can be regarded as a black and white image, so that the upsampling of the component design parameter space to the S -parameter matrix can be completed through transposed convolution.

The overall structure of the selected network used is illustrated in Fig. 3. The input design parameters are first expanded through the fully connected layer to enable better high-dimensional mapping and then transposed convolution is used to complete the task for predicting the S -parameter. The number of output channels of the last convolution layer is the sum of the number of $\text{Re}S_{ij}$ and $\text{Im}S_{ij}$ that need to be predicted. As for a two-port network, the output channels are $\text{Re}S_{11}, \text{Im}S_{11}, \text{Re}S_{21}$, and $\text{Im}S_{21}$, respectively, and the number is 4.

The number of RLNNs is determined by the number of subcomponents in the block. If the number of subcomponents in the block is h , h RLNNs are used as surrogate models for predicting S -parameters of each subcomponent, that is, $\text{RLNN}_1, \text{RLNN}_2, \text{RLNN}_3, \dots$, and RLNN_h . The effects of each network are as follows:

$$\tilde{S}_i = f_{\text{RLNN}_i}(\tilde{x}), \quad i = 1, 2, 3, \dots, h. \quad (4)$$

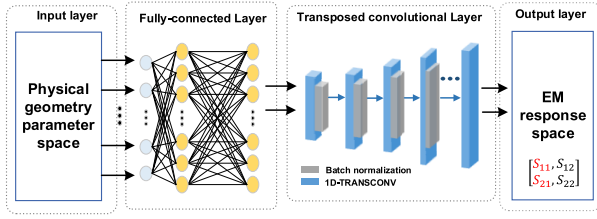


Fig. 3. Selected structure of the RLNN.

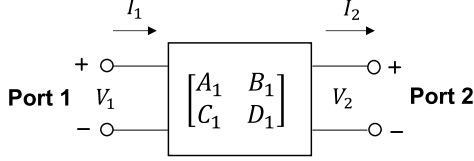


Fig. 4. Schematic of the two-port network.

After training, the network structure and weights of each RLNN will be preserved, which can facilitate the subsequent assembly of other complex structures composed of subcomponents corresponding to the RLNN.

C. S-Parameter Cascading Model

For the analysis of a microwave network, the S -parameter matrix provides a complete network description from the port of the N -port network. It links the voltage wave incident on the port with the reflected wave from the port. For some component networks, the scattering matrix can be directly calculated using network analysis technology, or the scattering parameters can be directly measured using a vector network analyzer. After obtaining the S -parameters, they can be converted into other matrices as needed, such as the impedance matrix, admittance matrix, and transmission ($ABCD$) matrix, as shown in Fig. 4 [42]. After obtaining S -parameter proxy models of each basic subcomponent through RLNNs, the S -parameters of the complex microwave network composed of these basic subcomponents can be obtained by cascading the S -parameters of each subcomponent in the order of basic assembly, which is achieved through the SCM.

For a two-port network, to achieve the cascade of S -parameters of multiple two-port networks, it is convenient to use the $ABCD$ matrix corresponding to each network. Since the conversion between the S -parameter matrix and $ABCD$ matrix is quite convenient, the $ABCD$ matrix can be regarded as a medium for the cascade of S -parameters. The $ABCD$ matrix of a two-port network can be defined using the total voltage and the total current

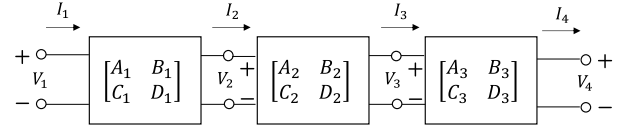
$$\begin{aligned} V_1 &= AV_2 + BI_2 \\ I_1 &= CV_2 + DI_2. \end{aligned} \quad (5)$$

The above equation in the form of a matrix can be written as

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}. \quad (6)$$

To facilitate the cascading of networks, I_2 is defined as the current flowing out of port 2 rather than the traditional current flowing into port 2. The cascaded two-port network is shown in Fig. 5. The cascaded two-port network can be expressed as follows:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} A_3 & B_3 \\ C_3 & D_3 \end{bmatrix} \begin{bmatrix} V_4 \\ I_4 \end{bmatrix}. \quad (7)$$

Fig. 5. Cascade of $ABCD$ matrices of multiple two-port networks.

The $ABCD$ matrix for the cascaded three networks is equal to the product of the $ABCD$ matrices of each two-port network. The order of the matrix product needs to be the same as the order of the two-port network, as matrix multiplication does not satisfy the commutative law. If ports 1 and 2 have the same port impedance Z_0 , the conversion of S -parameters to the $ABCD$ matrix and the conversion from the $ABCD$ matrix to the S -parameter matrix are detailed in [42].

If the numbers of both cascaded subelements and RLNNs used are h , the effect achieved by the SCM module is given as

$$\tilde{S}_{\text{casad}} = f_{\text{SCM}}(\tilde{S}_1, \tilde{S}_2, \tilde{S}_3, \dots, \tilde{S}_h) \quad (8)$$

where $\tilde{S}_{\text{casad}}$ is the cascaded S -parameter obtained by cascading the $ABCD$ matrices, and $\tilde{S}_i$ ($i = 1, 2, 3, \dots, h$) is the predicted output of the surrogate model of S -parameter corresponding to the h th subelements.

D. Coupling NN

In the design of microwave circuits, considering the mechanical or electrical transition between different media in the connection between different components, when the subcomponents are connected to form an overall structure, discontinuity effects will inevitably occur. This coupling effect at the connection between different components will cause a certain error between the S -parameters of the cascaded subcomponents and the S -parameters of the overall structure. Taking TSV-RDL interconnects as an example, the cascaded S -parameters are compared with the actual S -parameters of the overall structure, and the comparison results are intuitively shown in Fig. 6. It is evident that the cascaded S -parameters and the actual S -parameters are highly consistent in real and imaginary parts and amplitude, but there is a phase shift in the cascaded S -parameters, which is due to the influence of the coupling effect between different subcomponents.

To eliminate the influence of the coupling effect between different subcomponents, the CONN is proposed to achieve mapping from cascaded S -parameters to the overall S -parameter. Different from the low-dimensional to high-dimensional space mapping of geometrical parameters to the S -parameters, the mapping of cascaded S -parameters to the overall S -parameters is a mapping from high-dimensional space to high-dimensional space.

Note that the CNN can be used to learn the mapping from high-dimensional space to low-dimensional space, and the TCNN has a good effect in learning the mapping from low-dimensional space to high-dimensional space. Therefore, it is anticipated that the mapping effect from high-dimensional space to high-dimensional space can be well achieved by removing the fully connected layer in the traditional CNN and TCNN and then connecting the convolution layer of the CNN with the transposed convolution layer of the TCNN in series.

The CONN network structure is shown in Fig. 7. Another way to understand the effect of the CONN is that the convolution layer is used to realize the feature concentration and extraction of the cascaded S -parameters, so that

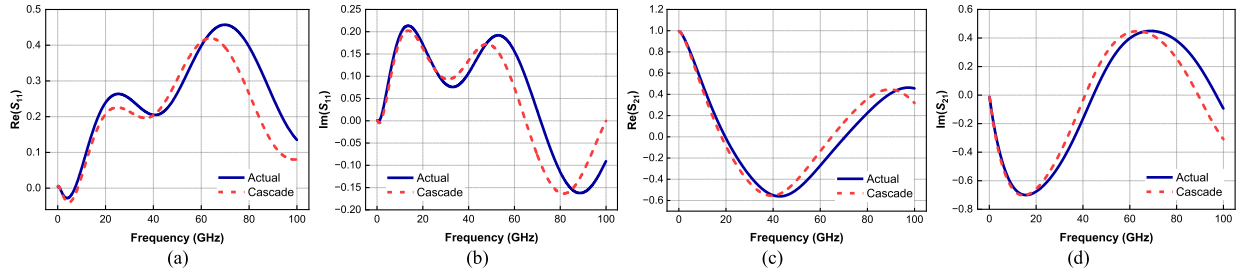


Fig. 6. Deviation between the coarse model and the fine model in predicting the S -parameters. (a) $\text{Re}(S_{11})$, (b) $\text{Im}(S_{11})$, (c) $\text{Re}(S_{21})$, and (d) $\text{Im}(S_{21})$.

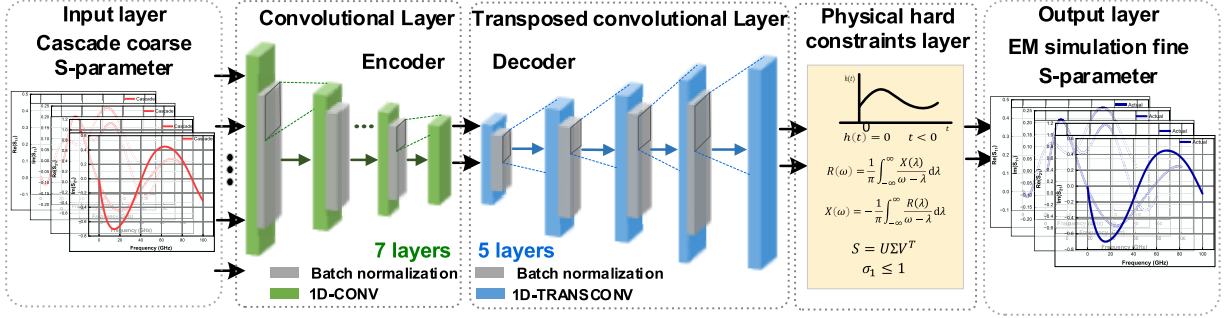


Fig. 7. Schematic of the CONN structure.

they can be represented by several low-dimensional features. This step can be regarded as an encoding process. The transposed convolution layer decodes the low-dimensional features extracted and encoded by the convolution layer and upsamples and maps them to the corresponding overall S -parameters. This step can be regarded as a decoding process. Through this encoder–decoder structure, the mapping from high-dimensional space to high-dimensional space is finally realized.

The CONN adopts an encoder–decoder architecture consisting of 1-D convolutional and transposed convolutional layers. The encoder part is composed of several 1-D convolutional (Conv1D) layers, which extract both local and global coupling features from the input cascaded S -parameter curves. Each layer maintains 30 channels while progressively reducing the temporal dimension. Batch normalization (BatchNorm) layers are applied to accelerate convergence and stabilize training. The decoder consists of five transposed convolutional layers, which gradually restore the output to the original curve resolution. BatchNorm is similarly used in the decoder. The final output of the decoder corresponds to the corrected S -parameters that account for real-world coupling effects.

Compared with the traditional method of directly mapping low-dimensional design parameter space to high-dimensional S -parameter space, the input of the CONN provides much more information to the output when mapping high-dimensional cascaded S -parameters to high-dimensional overall S -parameters, and, therefore, it can be anticipated that the CONN needs fewer datasets to achieve high accuracy. Moreover, when the same subcomponents are used to form other interconnect structures, the pretrained RLNNs are reused to generate the cascaded S -parameters, and accurate S -parameters can be obtained by training the CONN with a small amount of EM simulation and measurement datasets.

Although accurate S -parameters for various assemble structures can be flexibly and conveniently obtained using the proposed network architecture. The essence of the NN is a closed box, that is, the decision-making process of the algorithm is opaque and difficult to understand. All that can be

seen is that the input is put into the closed box, and the output is given by the box. In this process, the DNN contains millions of parameters and multiple layers of nonlinear operations, making it difficult even for researchers who design algorithms to track the specific logic behind each decision. When the network itself is learning and predicting S -parameters, it cannot learn its own physical properties. Therefore, the results obtained through the network should not violate the actual physical properties, that is, causality and passivity [43], [44]. For completion of the model and easy reference, the analysis of physical consistency is presented in the following, while a more detailed derivation is presented in [45].

1) *Causality Guarantee*: If a system H is a causal system, it can be defined as

$$h(t) = 0, \quad t < 0 \quad (9)$$

where $h(t)$ is the impulse response. For a causal system, its output depends only on current and past inputs, but not on future outputs. For an n -port system, its impulse response can be obtained by performing an inverse Fourier transform on the corresponding S_{ij} , to determine whether the S -parameters have caused causal violations. At the same time, for a causal system, the real and imaginary parts of its Fourier transform both constitute a Hilbert transform pair.

The Hilbert transform is an operation that shifts the phase of a signal by 90° . Unlike other transforms, such as the Fourier transform and Laplace transform, it is a transform of the same domain. Therefore, for a signal that satisfies causality, the imaginary part can be obtained by performing a Hilbert transform on its real part. The causal restriction based on S -parameter can be implemented using the Hilbert transform, that is, performing the Hilbert transform on the real part of the S -parameter predicted by NNs to obtain the imaginary part, so that the real and imaginary parts of the S -parameter predicted by the network constitute a Hilbert transform pair, thereby ensuring that the produced S -parameters satisfy causality. The Hilbert transform of the real part of the S -parameters is calculated by fast Fourier transform and inverse fast Fourier transform.

Algorithm 1 Causality Guarantee

Input: real part of S -parameter: S_{real} ; Number of frequency points: m

Output: S -parameter matrix with causality: S_{causal}

1. Construct the double-sided spectrum of S_{real} :

$$S_{\text{real}} = [S_{\text{real}}[0, \dots, m], S_{\text{real}}[m-1, \dots, 0]]$$

2. Perform a fast Fourier transform on S_{real} :

$$\mathcal{F} = \text{FFT}(S_{\text{real}})$$

3. Perform the following operations on $\mathcal{F}$:

$$\mathcal{Z} = \begin{cases} \mathcal{F}[0], & i = 0 \\ 0, & 1 \leq i \leq m+1 \\ 2\mathcal{F}[i], & m+1 \leq i \leq 2m-1 \end{cases}$$

4. Do IFFT on $\mathcal{Z}$ to obtain the real and imaginary part of the S -parameter:

$$S_{\text{real}} + jS_{\text{imag}} = \text{IFFT}(\mathcal{Z}_{\text{real}})$$

2) *Passivity Guarantee*: For the n -port passive component, the S -parameters of each sampling frequency point are a complex matrix S of size $n \times n$. The passivity criterion requires that the matrix of each sampling frequency point should satisfy the following properties:

$$\|S\|_2 \leq 1 \quad (10)$$

where $\|S\|_2$ is the two-norm of the matrix S . The two-norm of a square matrix is equal to its largest singular value

$$\|S\|_2 = \sqrt{\lambda_{\max}} = \sigma_{\max}. \quad (11)$$

The passivity of the matrix can be ensured by limiting the maximum singular value of the matrix to be less than or equal to 1. The singular value decomposition (SVD) algorithm will be used to directly perform SVD on the complex matrix of each frequency point of the S -parameters to achieve this operation.

There are many methods to use the maximum singular value of the S -parameter matrix to perform passivity constraints [46], [47]. Here, the passivity constraint is implemented by constructing a minimum phase filter, whose advantage is that the causality is not destroyed while ensuring the passivity. The specific causality and passivity guarantee algorithm is shown in Algorithms 1 and 2.

To ensure that the subcomponents and their assembled structures maintain physical consistency and can be directly used for rapid signal integrity analysis, the physical consistency hard constraints on S -parameters are implemented in the CONN. Hard constraints enforce physical or mathematical conditions directly within the model to ensure strict consistency, whereas soft regularization guides the model to satisfy these conditions through penalty terms [39].

III. DATA SCREENING METHODOLOGY BASED ON SA

The CONN is used to learn high-dimensional mapping from a cascaded S -parameter coarse model to the fine model. As the similarity between the coarse model and the fine model is high, the CONN can be regarded as a process of fine-tuning the accuracy of the curve. It can be easily found that the shape of the S -parameter curve will produce large changes in amplitude and the number of resonance points as the geometrical

Algorithm 2 Passivity Guarantee

Input: Complex S -parameter matrix for every frequency point: S ; Number of frequency points: m , number of ports: n

Output: S -parameter matrix with passivity: S_{passive}

For i in range(m):

1. Compute the transposed conjugate matrix $S^H(f_i)$ of $S(f_i)(f)$
2. Calculate the eigenvalues of the matrix $S^H(f_i)S(f_i)$:

$$\det(S^H(f_i)S(f_i) - \lambda I) = 0$$

3. Sort the eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ from large to small to obtain the eigenvalue matrix:

$$\lambda(f_i) = \begin{bmatrix} \lambda_1(f_i) & 0 & 0 & \dots & 0 \\ 0 & \lambda_2(f_i) & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n(f_i) \end{bmatrix}$$

4. Take the square root of each value of the eigenvalue matrix to get the singular value matrix:

$$\Sigma(f_i) = \begin{bmatrix} \sqrt{\lambda_1(f_i)} & 0 & 0 & \dots & 0 \\ 0 & \sqrt{\lambda_2(f_i)} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \sqrt{\lambda_n(f_i)} \end{bmatrix}$$

5. Obtain the max value with perturbation factor of S :

$$\check{\sigma}_1(f_i) = \sqrt{\lambda_1(f_i)}$$

6. Construct a minimum phase filter:

$$\Upsilon(f_i) = |\Upsilon(f_i)|e^{-j\psi(f_i)}$$

where

$$|\Upsilon(f_i)| = \begin{cases} \frac{1}{\check{\sigma}_1(f_i)}, & \check{\sigma}_1(f_i) > 1 \\ 1, & \check{\sigma}_1(f_i) \leq 1 \end{cases}$$

$$\psi(f_i) = \mathcal{H}\{\log|\Upsilon(f_i)|\}$$

$\mathcal{H}\{\}$ represents the Hilbert transform.

7. Use the minimum phase filter to ensure the passivity:

$$S_{\text{passive}}(f_i) = S(f_i) \odot \Upsilon(f_i)$$

parameters change. If the curve-type dataset learned by the CONN during the training is incomplete, it would be difficult to correct the curve types that are fewer in number or do not exist in the training set.

To ensure that the CONN training uses as few expensive fine datasets as possible, an SA method is developed by employing RLNNs to guide the complete selection of datasets for CONN training. SA is a method to evaluate the impact of each input on its output, and it generally consists of the following steps [48].

Step 1: Determine the input of the model and the sampling range.

Step 2: Generate the specific input parameters by running the sampling function.

Step 3: Input the generated input parameters into the model and save the output.

Step 4: Calculate the sensitivity indices by running the analysis function based on the output.

Regarding the completeness problem of CONN's training set selection, traditional SA methods often use the EM

simulation of the assembled structure as the model for analysis, thereby exploring the influences of geometrical parameters on the S -parameters. Since SA requires sampling of thousands of the model inputs, the EM simulation would be called each time when the model is run to generate the output, resulting in a large consumption of time and computing resources. Meanwhile, SA methods based on surrogate models have been demonstrated to achieve superior performance [49]. As RLNNs are used as surrogate models of subcomponents, their input geometrical parameters are subsets of the geometrical parameters of the whole structure. Here, the pretrained RLNNs, basic elements of the whole structure, are used for SA. Moreover, as shown in Fig. 6, the similarities of the S -parameters produced by the cascaded and the whole structures are high, and therefore, the SA of the RLNN is sufficient to find the important geometrical parameters. The Sobol SA is employed in this article [50]. It is a global SA method grounded in variance decomposition, enabling a quantitative assessment of the influence of input variables on the output. The model under investigation is partitioned into a single input variable and a function comprising multiple input variables.

The sampling frequency of the S -parameter curve can reach hundreds or thousands, and it contains much information, which is not convenient for analysis. As a commonly used frequency-domain analysis method, the wavelet transform is used to reduce the dimension and extract the information of S -parameters. The wavelet transform decomposes the signal into a linear combination of a set of basis functions. These basis functions are called “wavelets,” which are generated by a function called the mother wavelet through translation and scaling. In practical applications, the wavelet transform typically decomposes a signal into two components: approximate coefficients (ACs) and detail coefficients (DCs). ACs correspond to the low-frequency components, reflecting the overall trend of the signal. DCs correspond to the high-frequency components, capturing the finer details and rapid variations of the signal. Through multilevel decomposition, both low- and high-frequency characteristics of the signal at different scales can be obtained.

The specific procedure of SA is given as follows.

Step 1: Convert the real and imaginary parts of the output S_{11} and S_{21} by RLNNs into the form of $\text{dB}(S_{11})$ and $\text{dB}(S_{21})$.

Step 2: Apply the wavelet transform to two curves and obtain the approximate coefficient and detail coefficient energy.

Step 3: Calculate the sensitivity ranking of each input geometrical parameter for each output through the Sobol algorithm and mark the input with the highest sum of sensitivity values as the highest sensitivity input parameter of the RLNN.

Step 4: Generate a certain amount of dataset using Latin hypercube sampling (LHS) before training the CONN for the target assembled structure.

Step 5: Divide the training set and test set according to the highest sensitivity input parameters of the RLNNs, to ensure the distribution of the highest sensitivity input parameters in the training set is uniform.

IV. APPLICATION I: SICL-BGA INTERCONNECT STRUCTURE

The first application is an interconnect structure consisting of SICL and BGA [51]. SICL is a printed coaxial line inside the circuits, and it can be used as a chip-to-chip interconnect,

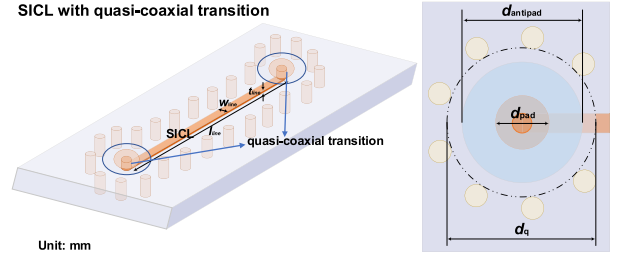


Fig. 8. 3-D schematic and top view of SICL with QCT.

TABLE I

PARAMETERS OF TWO BASIC COMPONENTS (UNIT: mm)

	Parameter	Symbol	Max	Min
SICL	Length	l_{line}	5	15
	Width	w_{line}	0.15	0.3
	Thickness	t_{line}	0.03	0.1
	Diameter pad	d_{pad}	0.5	0.9
	Diameter anti-pad	$d_{antipad}$	0.9	2.6
	Diameter QCT	d_q	1.1	2.8
BGA	Diameter pad	d_{pad}	0.5	0.9
	Diameter anti-pad	$d_{antipad}$	0.9	2.6
	Height	h	0.1	0.4

TABLE II

STRUCTURAL PARAMETERS OF THE RLNN AND CONN

	RLNN	CONN
Linear layers	4	0
Conv layers	0	7
T-conv layers	5	5
Channels of conv	0	4-30-30-30-30-30-50
Channels of T-conv		50-30-30-30-30
Strides of conv	0	2-1-2-1-4-2-2
Strides of T-conv	2-2-4-2-3	1-2-2-4-3
Activation Function	Relu	Relu
Hard physical constraints	✓	✓

as shown in Fig. 8. BGA is widely used in chip packaging. The quasicoaxial transition (QCT) can be used as a vertical interconnection to connect the SICL and BGA [52]. The proposed method is utilized for the precise modeling of the assembled interconnection structure, comprising SICL with QCT and BGA.

A. RLNN Model Settings and Training

The RLNN is used to learn the mapping from the geometrical parameters of the substructure to its corresponding S -parameter space. The number of RLNNs depends on the number of substructures. In this part, two RLNNs, named RLNN1 and RLNN2, are used to perform the modeling of the S -parameters of SICL and BGA. The geometrical parameter spaces of SICL and BGA are given in Table I. LHS is performed based on the ranges of geometrical parameters to obtain the input dataset of the RLNN, with 995 groups for each component. All input datasets, corresponding to each input group, are imported into the full-wave EM simulator ANSYS HFSS to obtain S -parameters from 0.1 to 30 GHz with a step size of 100 MHz. 796 sets of data in the dataset are used as training sets. The network structures, hyperparameters, and training strategies are adjusted to improve the accuracy. The final RLNN structure and CONN structure are summarized in Table II.

The Adam optimizer is adopted, and the learning rate is set to 0.003. There are 4000 iterations in total. The learning rate is reduced to half of the original value every 500 iterations,

TABLE III
SA RESULTS OF SICL

	d_{pad}	l_{line}	t_{line}	d_{pad}	$d_{antipad}$	d_q
AC(S_{11})	7.7%	4.9%	13.1%	57.8%	1.4%	5.8%
DC1(S_{11})	3.8%	42.0%	0.9%	-3.8%	1.8%	0.5%
DC2(S_{11})	7.3%	27.6%	2.6%	2.1%	4.5%	1.5%
AC(S_{21})	6.2%	2.0%	1.0%	4.0%	19.4%	49.8%
DC1(S_{21})	2.9%	22.4%	0.5%	0.9%	8.7%	10.9%
DC2(S_{21})	3.1%	13.3%	0.3%	1.7%	8.6%	18.6%
sum	31%	112%	18.4%	62.7%	44.4%	87.1%

TABLE IV
SA RESULTS OF BGA

	d_{pad}	h	$d_{antipad}$
AC(S_{11})	35.1%	0.1%	63.4%
DC1(S_{11})	8.8%	0.0%	67.7%
DC2(S_{11})	5.8%	2.4%	33.9%
AC(S_{21})	27.5%	0.5%	54.8%
DC1(S_{21})	14.6%	0.7%	51.6%
DC2(S_{21})	9.6%	2.9%	30.8%
sum	101.4%	6.6%	302.2%

and it no longer decreases at the 2500th generation. For the evaluation of the test set, the normalized mean squared error (NMSE) is selected as the evaluation indicator of the model. Compared with the mean squared error (mse), which is used to measure the difference between the predicted results and the actual results, the NMSE normalizes the difference to [1, 0] and is more suitable for comparing the prediction capabilities between different models. The proposed method is executed on a computer equipped with an i5-13500 CPU at 2.50 GHz and 32 GB of RAM, and the structure is simulated using commercial software ANSYS HFSS

$$NMSE = \frac{\sum ((y_{ture} - y_{pred})^2)}{\sum ((y_{true} - \text{mean}(y_{ture}))^2)}. \quad (12)$$

After training, the average NMSEs are 0.918% and 0.520% for RLNN1 and RLNN2, respectively. The corresponding training times are 20.95 and 18.88 min. The weights and biases of each RLNN are retained for the subsequent assemble process.

B. SA Based on RLNNs

Before the CONN training, the SA Sobol method based on RLNNs is used to guide the training set screening of the CONN. The real and imaginary parts of S_{11} and S_{21} output from the RLNN are transferred into $\text{dB}(S_{11})$ and $\text{dB}(S_{21})$. Then, the wavelet transform is performed on $\text{dB}(S_{11})$ and $\text{dB}(S_{21})$ to extract ACs and DCs. Here, Daubechies 4 is used as the wavelet basis function, and the level of wavelet transform is 2. The proposed SA method in Section III is applied to obtain the sensitivity of input geometrical parameters of SICL and BGA to their output S -parameters. The first-order sensitivity coefficients of SICL and BGA are shown in Tables III and IV, respectively.

An analysis of the RLNNs reveals the top n ($1 \leq n < m$, m represents the number of inputs analyzed for the RLNN) geometric parameters with the highest cumulative sensitivity for each substructure. In this case study, the high-sensitivity inputs selected for RLNN1, corresponding to the SICL structure, are the SICL length l_{line} and the quasioaxial diameter d_q . For RLNN2, corresponding to the BGA structure, the high-sensitivity input is the anti-pad diameter $d_{antipad}$.

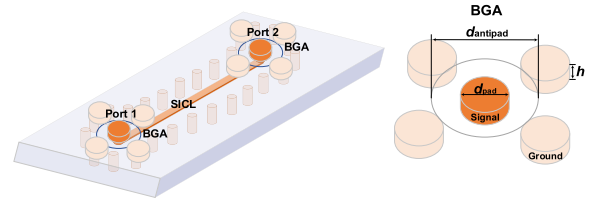


Fig. 9. Left: the BGA-SICL-BGA interconnection structure; Right: BGA.

TABLE V
BGA-SICL NMSEs OF THE PROPOSED METHODS WITH AND WITHOUT SA

Train	Test	Method	NMSE	Train time
53	141	TCNN	55.56%	2.45min
		TL-TCNN	41.60%	2.26min
		TL-FCNN	56.37%	1.45min
		Proposed (with SA)	13.55%	3.52min
		Proposed (without SA)	20.30%	3.67min
100	94	TCNN	15.05%	3.13min
		TL-TCNN	11.12%	3.01min
		TLFCNN	27.67%	1.88min
		Proposed (with SA)	7.27%	5.23min
		Proposed (without SA)	9.86%	5.15min
145	49	TCNN	11.98%	4.02min
		TL-TCNN	8.25%	3.79min
		TL-FCNN	22.56%	2.58min
		Proposed (with SA)	3.54%	7.35min
		Proposed (without SA)	8.10%	7.62min

C. Assemble Model Settings and Training

For the BGA-SICL-BGA interconnection structure named BGA-SICL, as shown in Fig. 9, 194 sets of geometrical parameters of the assemble structure are obtained through LHS. The EM simulator ANSYS HFSS is used to obtain 194 sets of geometrical parameters and corresponding EM response datasets for CONN training. Then, the geometrical parameters of each subcomponent corresponding to 194 groups of assemble structures are input into the RLNN with retained weights and biases, and the S -parameters are cascaded in the order of BGA-SICL-BGA through the SCM module. The obtained coarse model is used as the input of the CONN, and the refined results of the assemble structure parameters obtained through HFSS simulation are used as its output. Before training, the dataset is sorted in ascending order based on the highly sensitive inputs identified through SA. Subsequently, uniform interval sampling is performed for each sorted subset. The sampled data are then merged, and duplicate samples are removed to construct the final training dataset.

By adjusting the sampling interval, four training sets of sizes 30, 53, 100, and 145 are generated. Each training set is used for multiple training sessions, and the average test results are recorded. These results were compared to the average test results obtained by dividing the training set using different random seeds. To demonstrate the superiority of the proposed network on small-sample datasets. The detailed results are presented in Table V. Here, the training set is 145 groups, and the results are shown in Fig. 10.

A comparison is also made with the traditional method, directly modeling the S -parameter under the same dataset and transfer learning method. Traditional methods adopt the same parameter and hyperparameter settings, as well as training strategies, as the RLNN. Transfer learning builds upon an NN model pretrained on a source task to enhance generalization performance on a related target task through techniques such as feature freezing and fine-tuning. In this application,

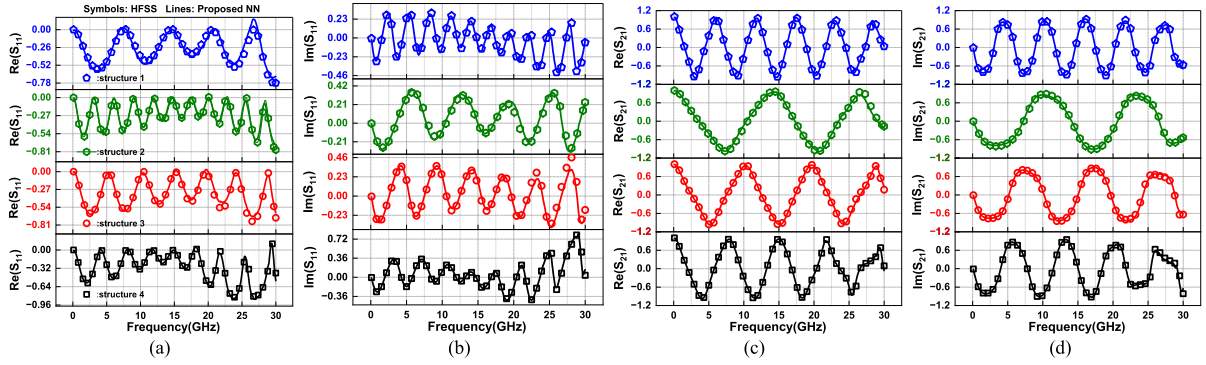


Fig. 10. Deviation between HFSS and the proposed NN with SA in predicting the S -parameters of BGA-SICL. (a) $\text{Re}(S_{11})$, (b) $\text{Im}(S_{11})$, (c) $\text{Re}(S_{21})$, and (d) $\text{Im}(S_{21})$.

the RLNN model corresponding to the SICL component (denoted as RLNN1) is used as the source model and transferred to the target task, which involves the modeling of the assembled interconnect structure.

To evaluate the effectiveness of transfer learning, two different NN architectures are adopted: a TCNN and a fully connected NN (FCNN). Their transfer learning counterparts are referred to as TL-TCNN and TL-FCNN, respectively.

For the TL-TCNN, the second, third, and fourth fully connected layers are frozen, and only the transposed convolutional layers are fine-tuned. For the TL-FCNN, which consists of five fully connected layers with 60 neurons each, the second and third layers are frozen, while the fourth and fifth layers are fine-tuned during transfer. The original FCNN trained on the SICL dataset (i.e., the source task) achieves an average NMSE of 3.20% with a training time of 22.82 min.

V. APPLICATION II: TSV-RDL INTERCONNECT STRUCTURE

The second application is interconnection structures consisting of TSV and RDL. The growing demand for high-speed signal transmission applications has brought about issues related to internal SI in 3-D integrated microsystems. TSV, as well as RDL, constitute the electrical interconnection for high-speed signal transmission in 3-D integrated microsystems. Based on the transmission line theory, an interconnect structure consisting of a pair of TSV-RDL in the ground-signal (G-S) transmission mode is defined as the basic channel within the system. In actual signal transmission, TSV and RDL can be used as the basic unit structure to form a variety of channel transmission structures, as shown in Fig. 11. Based on the assemble NN architecture proposed above, the RDL is first divided into two types according to its position relative to the silicon substrate. The RDL layer located above and below the silicon substrate is named RDL_{up} and RDL_{down} , respectively, and the RDL with a corner is named RDL_{corn} . RDL_{up} , RDL_{down} , RDL_{corn} , and TSV are used as the basic elements. The computational efficiency and configurations of the CONN, TCNN, and FCNN models are summarized in Table VI, where the training time is reported using the TSV-RDL1 case as a representative example.

A. RLNN Model Settings and Training

Here, four RLNNs are used to perform the modeling of S -parameters of RDL_{up} , RDL_{down} , RDL_{corn} , and TSV. The ranges of geometrical parameters of the four basic components are listed in Table VII. LHS is performed based on the ranges

	TCNN	FCNN	CONN
Parameter count	71004	499400	69454
FLOPs	5.889M	491.160K	20.553M
Training time	5.38 min	5.26 min	8.5 min
Inference time	2.174 ms	0.905 ms	5.204 ms

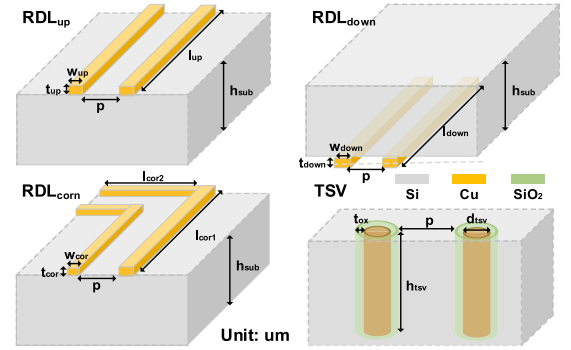


Fig. 11. Basic components: RDL_{up} , RDL_{down} , RDL_{corn} , and TSV.

	Parameter	Symbol	Max	Min
RDL_{up}	Length	l_{up}	100	700
	Width	w_{up}	10	30
	Thickness	t_{up}	2	5
	Pitch	p	40	60
	Substrate height	h_{sub}	56	106
RDL_{down}	Length	l_{down}	100	700
	Width	w_{down}	10	30
	Thickness	t_{down}	2	5
	Pitch	p	40	60
	Substrate height	h_{sub}	56	106
RDL_{corn}	Length1	l_{cor1}	100	700
	Length2	l_{cor2}	100	700
	Width	w_{cor}	10	30
	Thickness	t_{cor}	2	5
	Pitch	p	40	60
	Substrate height	h_{sub}	56	106
TSV	Height	h_{TSV}	50	100
	Diameter	d_{TSV}	10	30
	Pitch	p	40	60

of geometrical parameters to obtain the input dataset of the RLNN, with 1000 groups for each component. All input

TABLE VIII
TOP THREE SA RESULTS OF ALL SUBSTRUCTURES

RDL _{up}		RDL _{down}		RDL _{corn}		TSV	
l_{up}	237%	l_{down}	251%	w_{cor}	126%	d_{TSV}	133%
w_{up}	80%	w_{down}	111%	l_{cor1}	80%	h_{TSV}	122%
p	29%	p	42%	l_{cor2}	70%	p	43%

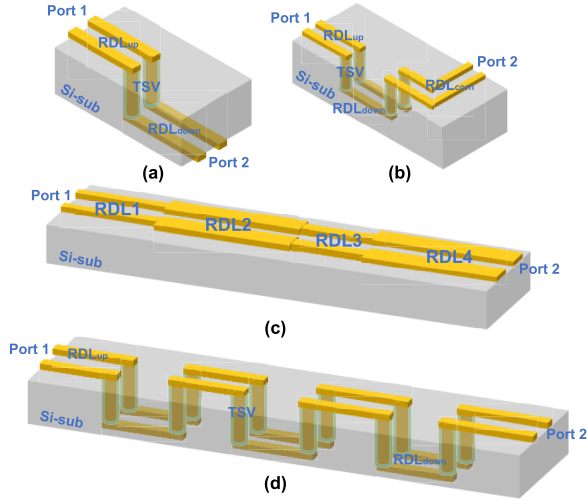


Fig. 12. Two assemble structure consists of TSV and RDL. (a) TSV-RDL1, (b) TSV-RDL2. (c) RDL-discontinuity. (d) TSV-RDL3.

datasets are imported into the full-wave EM simulator ANSYS HFSS to obtain S -parameters with a step size of 100 MHz from 0.1 to 100 GHz corresponding to each input group. After all simulated results are obtained, four RLNNs, that is, RLNN1, RLNN2, RLNN3, and RLNN4, are trained based on the datasets corresponding to RDL_{up}, RDL_{down}, RDL_{corn}, and TSV, respectively. 800 sets of data in the dataset are used as training sets.

After training, the average NMSEs are 0.828%, 0.710%, 0.709%, and 0.234% for RLNN1, RLNN2, RLNN3, and RLNN4, respectively. The corresponding training times are 36.17, 35.80, 37.23, and 35.31 min. The weights and biases of each RLNN are retained for the subsequent assemble process.

B. SA Based on RLNNs

For this application, the same SA method is conducted on four RLNNs, each representing one of the following structures: RDL_{up}, RDL_{down}, RDL_{corn}, and TSV. The top three sum first-order sensitivity coefficients of RDL_{up}, RDL_{down}, RDL_{corn}, and TSV are shown in Table VIII. In this application, the high-sensitivity inputs selected are l_{up} for RDL_{up}, l_{down} for RDL_{down}, w_{cor} for RDL_{corn}, and h_{TSV} and d_{TSV} for TSV.

C. Assemble Model Settings and Training

Here, the proposed assemble NN is used to model assemble structures as shown in Fig. 12. For the RDL_{up}-TSV-RDL_{down} interconnect structure, which is named TSV-RDL1, 200 sets of geometrical parameters of the assemble structure are obtained through LHS. The geometrical parameters and corresponding EM response datasets are obtained through EM simulation for CONN training. Subsequently, the geometric parameters of each subcomponent are input into their corresponding RLNNs, and the SCM module is employed to obtain the cascaded S -parameter coarse model. The obtained coarse model is used as the input of the CONN, and the refined results of

TABLE IX
TSV-RDL1 NMSEs OF THE PROPOSED METHODS WITH AND WITHOUT SA

Train	Test	Method	NMSE	Train time
34	166	TCNN	14.22%	2.68min
		TL-TCNN	13.65%	2.70min
		TL-FCNN	12.21%	2.59min
		Proposed (with SA)	2.10%	4.01min
		Proposed (without SA)	5.18%	4.22min
74	126	TCNN	10.65%	4.25min
		TL-TCNN	4.85%	4.31min
		TL-FCNN	5.20%	4.28min
		Proposed (with SA)	0.95%	6.12min
		Proposed (without SA)	2.51%	6.43min
102	98	TCNN	8.76%	5.45min
		TL-TCNN	4.30%	5.38min
		TL-FCNN	3.61%	5.26min
		Proposed (with SA)	0.66%	8.70min
		Proposed (without SA)	2.05%	8.55min

the assemble structure parameters obtained through HFSS simulation are used as its output.

Before the training, the high-sensitivity inputs selected are used to guide the training set screening. To verify the performance of the CONN and SA, the number of training sets is gradually increased during training. The results trained using the proposed method with and without SA are compared with the traditional method of directly mapping geometrical parameters to S -parameter networks, TL-TCNN and TL-FCNN. The RLNN model corresponding to the RDL_{up} component is used as the source model of the TL-TCNN. The comparison between proposed methods with and without SA is conducted under identical settings, with the only variable being the screening strategy to show the performance of the SA method. It should be noticed that while the original FCNN trained on the RDL_{up} dataset achieves an average NMSE of 0.435% with a training time of 40.71 min. The average NMSE comparison results are shown in Table IX.

As the CONN training set increases to 102 samples, four representative test results are shown in Fig. 13. The model achieves high accuracy even with limited training data, while maintaining good physical consistency in the S -parameters. For the second assembled structure, TSV-RDL2—where an additional connection between TSV and RDL_{corn} is introduced—CONN is trained using the same procedure as in the first case and compared with direct geometry-to- S -parameter networks, TL-TCNN, and TL-FCNN. Results are summarized in Table X, with four test cases illustrated in Fig. 14 when trained on 150 samples. The third structure, RDL-discontinuity, involves a three-step discontinuous RDL. Again, the CONN is trained similarly and compared with baseline methods, with results in Table XI and test cases shown in Fig. 15 (121 training samples). Lastly, TSV-RDL3 introduces multiple connections among RDL_{down}, TSV, and RDL_{up}. Following the same training strategy, the CONN is compared against other models, as shown in Table XII and Fig. 16 (120 training samples).

As shown through comparative analysis across various application scenarios, the proposed architecture leverages a knowledge transfer mechanism, where a coarse-to-fine mapping is constructed using the cascaded S -parameter model derived from submodules. This enables high modeling accuracy even with limited training data, significantly reducing the need for costly EM simulations. Although the CONN requires slightly more training time than baseline methods,

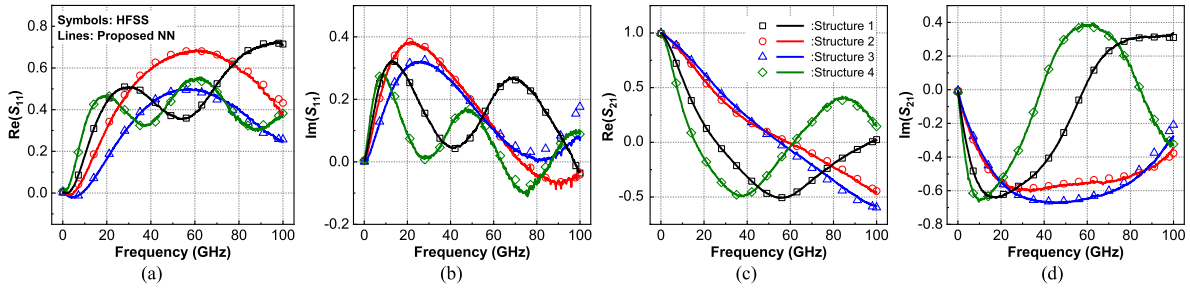


Fig. 13. Deviation between HFSS and the proposed NN with SA in predicting the S -parameters of TSV-RDL1. (a) $\text{Re}(S_{11})$, (b) $\text{Im}(S_{11})$, (c) $\text{Re}(S_{21})$, and (d) $\text{Im}(S_{21})$.

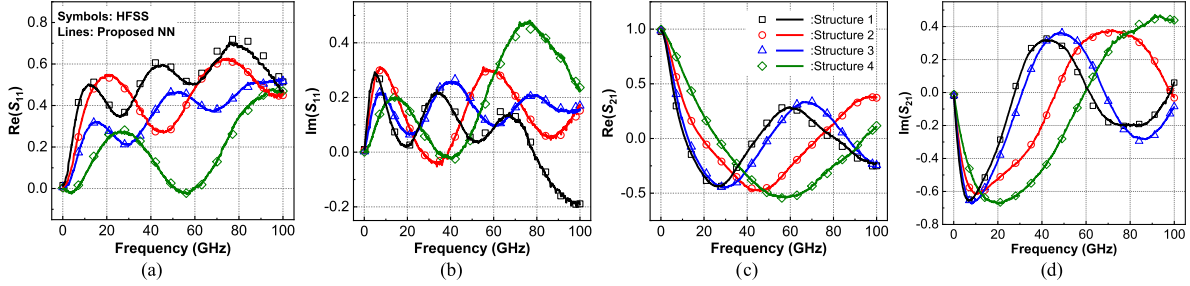


Fig. 14. Deviation between HFSS and the proposed NN with SA in predicting the S -parameters of TSV-RDL2. (a) $\text{Re}(S_{11})$, (b) $\text{Im}(S_{11})$, (c) $\text{Re}(S_{21})$, and (d) $\text{Im}(S_{21})$.

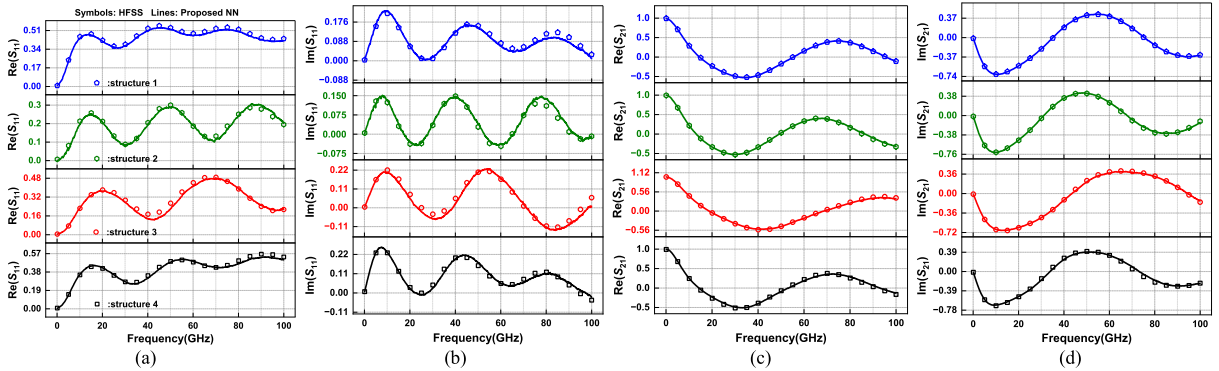


Fig. 15. Deviation between HFSS and the proposed NN with SA in predicting the S -parameters of RDL-discontinuity. (a) $\text{Re}(S_{11})$, (b) $\text{Im}(S_{11})$, (c) $\text{Re}(S_{21})$, and (d) $\text{Im}(S_{21})$.

TABLE X

TSV-RDL2 NMSES OF THE PROPOSED METHODS WITH AND WITHOUT SA

Train	Test	Method	NMSE	Train time
50	142	TCNN	20.24%	3.83min
		TL-TCNN	19.43%	3.30min
		TL-FCNN	19.25%	3.00min
		Proposed (with SA)	3.12%	4.15min
		Proposed (without SA)	9.00%	4.28min
96	66	TCNN	15.83%	4.89min
		TL-TCNN	15.43%	4.80min
		TL-FCNN	16.24%	4.86min
		Proposed (with SA)	2.91%	7.43min
		Proposed (without SA)	5.03%	7.29min
150	42	TCNN	11.61%	7.79min
		TL-TCNN	9.65%	7.01min
		TL-FCNN	10.18%	7.30min
		Proposed (with SA)	1.29%	11.23min
		Proposed (without SA)	2.02%	10.89min

TABLE XI

RDL-DISCONTINUITY NMSES OF THE PROPOSED METHODS WITH AND WITHOUT SA

Train	Test	Method	NMSE	Train time
54	147	TCNN	46.53%	3.58min
		TL-TCNN	44.89%	3.61min
		TL-FCNN	44.35%	3.86min
		Proposed (with SA)	3.63%	4.75min
		Proposed (without SA)	5.99%	4.90min
102	99	TCNN	10.38%	5.51min
		TL-TCNN	9.88%	5.53min
		TL-FCNN	10.48%	5.22min
		Proposed (with SA)	3.11%	7.51min
		Proposed (without SA)	4.83%	7.56min
121	80	TCNN	8.91%	5.81min
		TL-TCNN	7.43%	5.92min
		TL-FCNN	9.10%	6.74min
		Proposed (with SA)	2.14%	9.77min
		Proposed (without SA)	3.19%	8.65min

this overhead is justified by its efficiency gains. Additionally, since RLNNs are pretrained and reusable, the computational cost of applying the method repeatedly remains minimal.

In contrast, conventional methods that directly map geometric parameters to S -parameters cannot incorporate prior knowledge, making them less effective in small-data regimes.

Transfer learning addresses this by reusing knowledge across tasks but suffers from limitations such as the absence of task similarity metrics, reliance on deep features, and reduced interpretability. Moreover, traditional transfer learning typically transfers knowledge between single networks, which only allows partial reuse of substructure knowledge in the

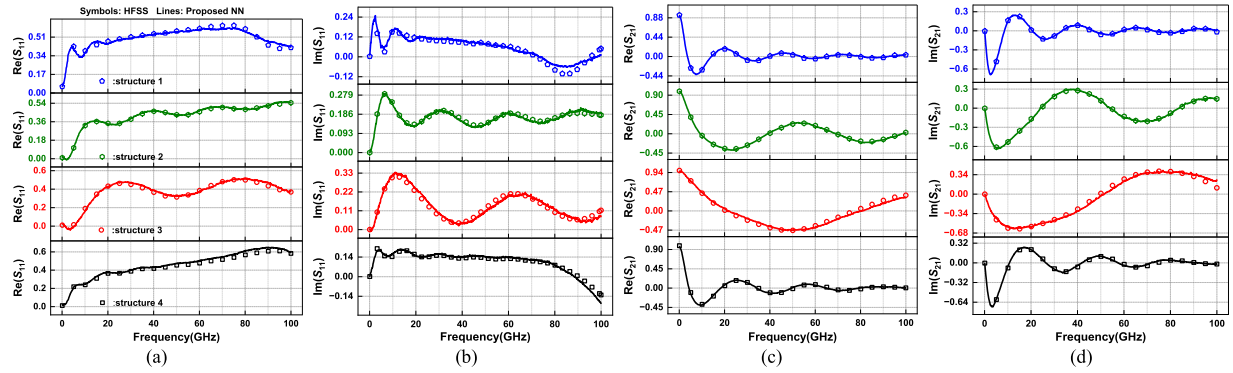


Fig. 16. Deviation between HFSS and the proposed NN with SA in predicting the S -parameters of TSV-RDL3. (a) $\text{Re}(S_{11})$, (b) $\text{Im}(S_{11})$, (c) $\text{Re}(S_{21})$, and (d) $\text{Im}(S_{21})$.

TABLE XII
TSV-RDL3 NMSEs OF THE PROPOSED METHODS
WITH AND WITHOUT SA

Train	Test	Method	NMSE	Train time
53	125	TCNN	29.20%	3.95min
		TL-TCNN	24.92%	3.54min
		TL-FCNN	28.60%	4.29min
		Proposed (with SA)	12.25%	4.84min
		Proposed (without SA)	15.23%	4.66min
76	102	TCNN	20.75%	5.70min
		TL-TCNN	22.20%	4.18min
		TL-FCNN	19.73%	4.05min
		Proposed (with SA)	8.32%	6.19min
		Proposed (without SA)	11.48%	6.30min
120	58	TCNN	11.92%	6.35min
		TL-TCNN	10.41%	5.81min
		TL-FCNN	13.69%	5.99min
		Proposed (with SA)	3.17%	9.18min
		Proposed (without SA)	9.30%	9.23min

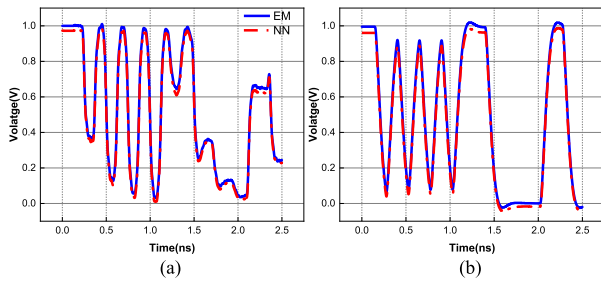


Fig. 17. Time-domain waveforms obtained by EM simulation and the proposed NN. (a) SICL-BGA and (b) TSV-RDL.

assembled interconnect modeling scenario. In contrast, the proposed method enables systematic reuse of all component models, offering more comprehensive and task-aligned knowledge transfer. To verify the engineering applicability of the proposed method, results from the assembled NN models for SICL-BGA and TSV-RDL1 structures are compared with HFSS simulations, with frequency-domain data converted to time-domain waveforms in ADS. As shown in Fig. 17, the method exhibits strong consistency with full-wave simulations, confirming its suitability for engineering applications.

VI. DISCUSSION

While the proposed assemble NN framework (CONN) demonstrates strong performance in modeling linear, passive interconnect structures, its modular design and encoder-decoder architecture also offer promising potential for future extensions to more complex interconnect behaviors.

One important direction lies in extending the modeling capability to nonlinear and time-varying interconnect systems. This may be achieved by transforming the modeling domain from frequency to time, allowing the network to learn nonlinear distortion and dynamic variations directly from waveform responses. Incorporating temporal modeling components such as the LSTM, GRU, or attention mechanisms may further enhance the ability to capture dynamic features in such systems.

Several limitations of the current framework should be noted. First, its accuracy depends on the quality of component-level surrogate models (RLNNs); poorly trained RLNNs can introduce cumulative errors. Second, in large-scale or irregular topologies—such as asymmetric crosstalk or multipath feedback—the local S -parameter cascading assumption may yield inaccurate coupling representations. Third, while the data demand is lower than that of end-to-end models, high-fidelity results still require sufficient sampling density and parameter coverage.

Additional challenges arise when dealing with special interconnect structures or extreme conditions. For example, modeling heterogeneous systems with varying geometric scales (e.g., microbump-RDL-substrate in 3-D packaging) requires finer spatial resolution and more precise coupling representation. Extending CONN to multiport differential channels introduces scalability issues due to increased coupling complexity and port asymmetry. At subterahertz frequencies (>100 GHz), rapid phase oscillations become harder to capture, potentially requiring deeper models or phase-aware constraints.

To address these challenges, future work may focus on the following.

- 1) Multiscale learning frameworks with mixed-resolution modules to jointly model local and global interconnect effects.
- 2) Phase-aware regularization to improve sensitivity to high-frequency oscillations.
- 3) Topology-guided encodings (e.g., graph-based or positional) to enhance structural generalization in complex configurations.

These advancements will further establish the proposed method as a versatile, interpretable, and reusable framework for next-generation signal integrity modeling.

VII. CONCLUSION

In this article, a novel multi-NN architecture is proposed for predicting the S -parameters of assembled interconnect-

tion structures. The assembled system is first decomposed into basic substructures, each modeled by a reusable RLNN to accurately capture individual EM behavior. The overall coarse S -parameter model is obtained by cascading RLNN outputs based on the assembly sequence. A coupling-aware encoder-decoder network (CONN) is then introduced to map the coarse model to a refined one using only a small training dataset. To further enhance CONN's performance on limited data, a sensitivity-based data screening method is applied to optimize training set selection, improving efficiency and reducing reliance on expensive EM simulations. Physical consistency constraints are enforced in each NN, ensuring that all predicted S -parameters are usable for downstream analysis. Compared with conventional transfer learning methods, the proposed approach enables more effective reuse of pretrained model knowledge in the current application scenarios. The method's effectiveness is demonstrated in two applications: Application I validates its accuracy in modeling complex multipole S -parameter curves, while Application II highlights its flexibility across different assembly configurations.

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